Journal of Engineering Science and Technology Review 18 (1) (2025) 235-241

Research Article

JOURNAL OF
Engineering Science and Technology Review

www.jestr.org

Principles and Challenges of Terahertz Wireless Networking System

A. Sammaiah* and M. Pranay Kumar

Electronics and Communication Engineering, Chaitanya Deemed to be University Hanamkonda, Telangana-506001, India.

Received 2 December 2024; Accepted 29 January 2025

Abstract

In order to achieve ultra-high speed and massive capacity, THz communication has become one of the significant technical options for 6th Generation. This paper focuses on the purpose and performance requirement of THz(terahertz) communication in various scenarios such as indoors and outdoors, and based on the needs of THz wireless networks, it details the new research development of key core technology such as network topologies, Media Access Control (MAC) layer protocol and neighbor node finding. Finally, it discusses the challenges faced by THz wireless networking and describes possible upcoming research directions.

Keywords: RDLG, Sleep, Distillation, Reinforcement learning

1. Introduction

As the commercialization of 5G mobile communications accelerates, countries around the world have gradually started research on next-generation mobile communication technologies, such as AI-enabled 6G intelligent access network technology and intent-driven network technology [1]. In the current sub-6 GHz and 5G millimeter wave frequency bands, these technologies can only achieve communication rates of the order of Gbit/s [2], which cannot meet the 6G rate requirements. In the future, the rates of local area networks and personal area networks will reach tens of Gbit/s; the minimum rate for smooth operation of virtual reality (VR) is as high as 10 Gbit/s; At present, core communication hardware such as THz huge-power devices and large scale arrays still have defects, which makes visible light have the advantage of wider coverage in indoor scenes. Compared with visible light communication, terahertz can achieve point-to-multipoint coverage, penetrate clouds and dust, and has strong anti-interference ability. Therefore, terahertz indoor multi-point access and outdoor communication technology have become research hotspots at home and abroad.

In 2013, the Defense Advanced Research Projects Agency (DAPRA) of the United States Department of Defense achieved the world's highest-speed terahertz communication within a range of 0.5 m, with a carrier frequency of 100 GHz and a rate of 200 Gbit/s [3]. In 2017, the China Academy of Engineering Physics achieved the world's longest-distance terahertz communication, with a carrier frequency of 140 GHz, a distance of 21 km, and a rate of 5 Gbit/s [4]. However, due to the low transmission power of terahertz radio frequency devices and the low gain of large-scale antenna arrays, these studies cannot support massive multiple-input multiple-output (Massive MIMO) communications [5], making new wireless networking technologies for terahertz narrow-beam directional antennas a hot topic and difficulty in current research.

This paper first introduces the application scenarios and requirements of terahertz communication, including indoor networking and outdoor networking. Then, based on the scenario requirement, the research status of THz wireless network is reviewed from the perspective of key narrow beam communication technology such as topologies, MAC layer protocol and neighbour node discovery. Finally, the problems and challenges currently faced by THz wireless networks are analyzing and discussing, and possible future research directions are describe.

2. THz Wireless networking application scenarios

Table 1 shows the performance requirements for THz wireless network. THz wireless network is divide into two scenarios: Indoor & Outdoor. These differ in terms of coverage, data rate, connectivity, latency and BER. These differences mainly depend on the depth of 6G network node connections: 6G wireless communication networks are oriented to all services, and a more open network architecture is determined for different networking scenarios to achieve the integration of communication, perception, and computing capabilities. On this basis, the connection of everything and the interactive performance constitute the deep connection of network nodes [6].

2.1 Indoor Application Scenarios

Terahertz wireless local area networks (TLAN), terahertz wireless personal area networks (TPAN), and information showers (IS) are the three types of indoor terahertz wireless networking scenarios that may be separated based on the extent of coverage.

Figure 1 displays the TLAN and TPAN situations. The rate is up to 100 Gbit/s, and the coverage criteria are less than 50 m and less than 20 m, respectively. By using a sub-array antenna structure, the network access point (AP) can transmit data to multiple users in different directions simultaneously, achieving high-speed communication between various network access devices (like PCs, mobile

*E-mail address: samkitsw@gmail.com
ISSN: 1791-2377 © 2025 School of Science, DUTH. All rights reserved.
doi:10.25103/jestr.181.24

terminals, and smart wearable devices) in contrast to traditional low-frequency WLAN. IS refers to short-distance Tbit/s communication within a coverage range of less than 5 m while meeting certain mobility requirements, mainly for pre-extraction of data such as files and video streams [7].

Table 1. Performance requirements of THz wireless networking

Category	Scenario	Cover	Rate	Latency	Connectivity	Target BER
Indoor Applications	Information Shower [7]	0.1~5m	1 Tbit/s	Seconds	Point-to-point, point-to-multipoint	
	Terahertz LAN	<60m	Up to 100Gbit/s	<1 ms	Point-to-point, point-to-multipoint, Ad Hoc	10^{-6}
	Terhertz personal area network [8]	<30m	>100Gbit/s	<1 ms	point-to-multipoint, Ad Hoc	10^{-6}
Outdoor Applications	Micro Ultra-Dense Cellular Network	10~16m	>100 Gbit/s	Millisecond	Point-to-multipoint	-
	Internet of Vehicles and Autonomous Driving [9]	>100m	>100 Gbit/s	Millisecond	Point-to-point, point-to-multipoint, Ad Hoc	-
	Military Communications	>100m	10~100Gbit/s	Millisecond	Point-to-point, point-to-multipoint, Ad Hoc	-
	Space Networking [10]	Km level	10~100Gbit/s	Seconds	Point-to-point, point-to-multipoint	-

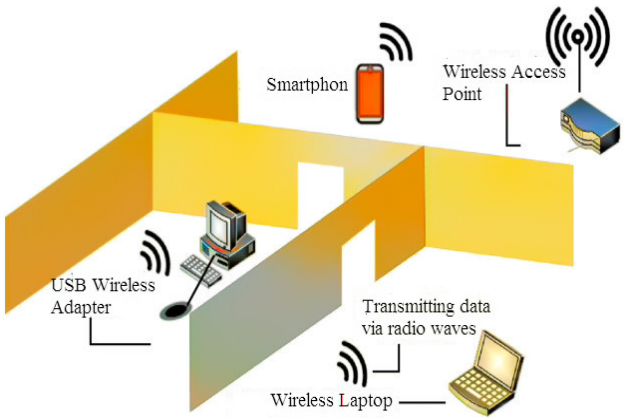

Fig. 1. TPAN and TLAN scenarios [11].

Terahertz wireless networking can provide high-speed and mobility guarantees for indoor communications, significantly improving the transmission capacity of high-quality indoor videos, such as high-definition holographic video conferencing, ultra-high-resolution video file downloads, and VR technology applications.

2.2 Outdoor Application Scenarios

Outdoor terahertz wireless networking scenarios are very broad, including ultra-dense microcells, Internet of Vehicles, military communications, and space communications.

The coverage scenario of terahertz cellular network is shown in Figure 2. The terahertz access point (THz-AP) covers dense microcells with a coverage range of 10 to 15 m, provide large-scale high-definition live broadcasts, 3D holographic communications and various services to mobile users in hot spots such as stadiums, concerts and tourist attractions. It can be foreseen that due to the large terahertz path loss, dense network nodes in hot spots and high mobility, terahertz ultra-dense microcells still need to use large-scale antenna arrays or other new network equipment to support network mobile communications. In order to overcome the problems caused by user mobility and non-line-of-sight transmission, there are two directions for the development of terahertz cellular networks in the future: with the help of smart antennas and reflectors, the reflected terahertz signals are directed to areas with high user density; with low-frequency fusion networking, macro base stations are used to achieve low-frequency coverage, and terahertz directional links are used to achieve high-speed data service transmission.

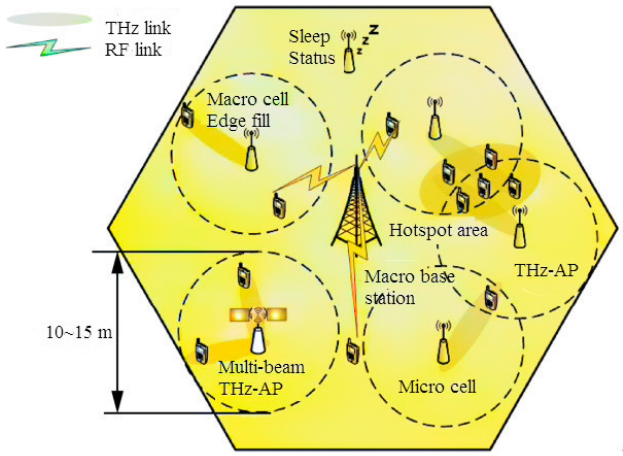

Fig. 2. Cellular network of THz.

The terahertz vehicle networking communication scenario is shown in Figure 3. The network coverage range is required to be greater than 100 m. As a distributed network for 6G, in addition to the high speed and low latency requirements, the terahertz vehicle networking also needs to achieve long-distance communication and high-speed vehicle mobility management, and it is urgent to improve network performance through terahertz networking. Vehicles need to process complex map and road information during driving, According to the business type, the terahertz vehicle networking can be divided into vehicle-to-vehicle (V2V) scenario and vehicle-to-infrastructure (V2I) scenario. In the V2V scenario, adjacent vehicles can share perception information through terahertz links [12]. Vehicles use shared information to establish a satellite view of the surrounding traffic conditions, display hidden objects in blind spots, and avoid collisions with other vehicles and people. In the V2I scenario, infrastructure units can collect sensor data about vehicles and surrounding traffic, and their communication and backhaul link performance can be improved through terahertz communication. Google's self-driving cars make sensor information at a rate of 760 Mbit/s and about 1.5 TB

of data in a single trip [13]. Transmitting and processing this data requires an ultra-high-speed communication network through terahertz networking to improve communication efficiency and reduce network latency.

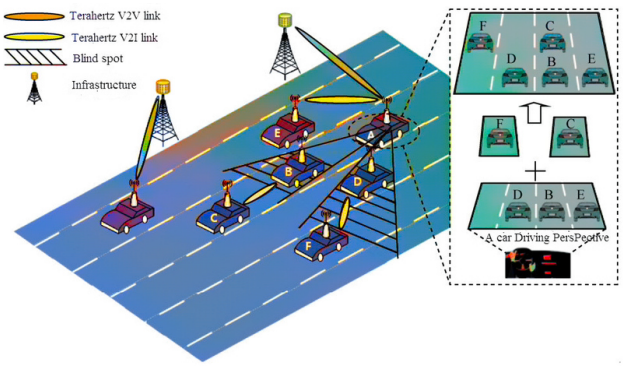

Fig. 3.THz vehicle networking scenario

The terahertz military communication scenario is shown in Figure 4. The coverage requirement is greater than 100 m, and the transmission rate is 10 to 100 Gbit/s. In modern warfare, unmanned aerial vehicles (UAVs) replace human pilots in combat and reconnaissance situations to perform a series of dangerous actions [14] and form battlefield command networks. The high-speed and large-capacity characteristics of terahertz will bear the 'UAV' networking to transmit high description lossless video to the combat command centre in real time, helping combat units to better analyze the battlefield situation. In addition, on the battlefield and in harsh environments, military vehicles and aircraft need to transmit a large amount of confidential data to assist other network devices in carrying out attacks or reconnaissance. The narrow beam characteristics of terahertz can ensure the security of this data.

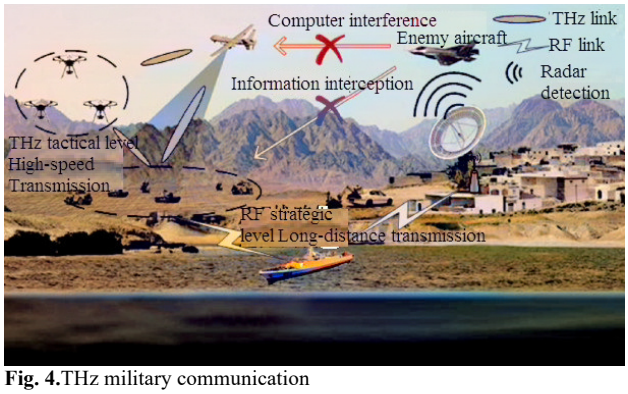

Fig. 4.THz military communication

As the core scenario of the 6G "ubiquitous coverage" communication network, the integration of space-ground-air has been the focus of research in the industry [15-16]. Terahertz can penetrate the atmosphere and the plasma sheath wrapped around the outside of high- speed aircraft, and is suitable for high-speed communication between satellites and between satellites and the ground. Compared with millimeter waves, terahertz has a larger bandwidth and faster transmission rate; compared with lasers, terahertz space links have less attenuation.

3. Research status of THz wireless networking

THz wireless network mostly focus on three key technologies: topologies control, MAC design, and neighbor node discovery [17]. The current status of terahertz networking research is shown in Table 2.

3.1 Topology Control

3.1.1 Centralized topology

The benefit of centralized topology is that Access Point (such as base station) can take advantage of centralized processing, efficiently assign network resource and realize multi user schedule, thus improving the overall performance of the network. Reference [18] proposed a centralized topology using terahertz links. In this network, AP uses directional antennas for node discovery scanning, initial link access and data transmission; user nodes first use omnidirectional antennas to establish efficient association with APs, and then switch to directional antennas for high-speed data transmission. In terms of access, the document discusses a beam switching access technology that can periodically perform beam alignment during link initialization access and information transmission, effectively reducing the delay caused by switching between omnidirectional antennas and directional antennas. In addition, the document also proposes to use local storage to improve AP scanning efficiency, enable AP to quickly establish connection with user nodes and schedule data transmission between nodes.

3.1.2 Distributed Topology

The distributed topology network structure has the characteristics of huge flexibility and sturdy self-organization ability, and is one of the application topology models of terahertz wireless networking in the future. In the distributed terahertz wireless network scenario, the time-varying nature of the network will cause frequent changes in the network topology, which requires corresponding control protocols to manage. Reference [9] considers switching the communication frequency based on the distance based on software defined network (SDN), providing a new idea for terahertz wireless networking, namely, microwave, millimeter wave, and terahertz wave fusion coverage and channel access; Reference [19] proposes a distributed topology in which directional antennas and omnidirectional antennas are used in coordination, with control signals using 2.4 GHz links and data transmission using terahertz communication links. However, these multi-band communication schemes will increase equipment costs and switching delays [20]. Reference [21] proposed an intelligent and safe spectrum control strategy for different indoor access subnets, and the nodes achieve anti-interference by adaptively selecting sub-frequency. However, these topologies face the problem of access deafness caused by narrow beams. Reference [22] proposed a wireless network with a high-speed terahertz communication link, in which the network nodes are distributed in a circular range of 10 m. The handshake protocol is initiated by the receiving end to reduce the network delay and packet loss rate..

3.2 Media Access Control Layer Protocol Design

The MAC(Media Access Control) layer control the communication among the physical layer hardware and the wireless transmission medium. In 2008, IEEE began to conduct standardization research on terahertz communications, and established the 100 GHz Wireless Task Group (TG100 GHz) in 2014, proposing the first terahertz band wireless communication standard IEEE Std.802.15.3d-2017. The standard was officially approved

for use in 2017, with a frequency band covering 253 to 322 GHz and a bandwidth of IEEE 69 GHz. However, unlike IEEE 802.11 and other WLAN protocols.

In addition to standardized protocols, research on MAC layer protocols of THz network can be divided into random multiple access, fixed multiple access and hybrid multiple access according to the multiple access mechanism.

Table 2. Current status of THz networking

Year	References	Frequency band	Topology	Multiple access mode	Simulation Tools	Performance Indicators
2017	[8]	340 GHz	Centralized	mix	OPNet	Latency, throughput
	[18]	0.06~10THz	Centralized	Fixed multi band	custom	data rate, latency, outage probability
	[19]	2.4GHz~0.1THz	distributed	Random, multi-band	Monte Carlo	latency, outage probability
	[20]	1.024 THz	distributed	CSMA	custom	Throughput, optimal distance
2021	[22]	1.04 THz	distributed	CSMA	NS-3	Packet transmission success probability, data delay, and throughput

(1) The benefit of random multiple access is that it has low control overhead. Reference [20] considered the cross-layer impact of terahertz channels and directional antennas on different layers based on CSMA, and studied the relay distance to achieve the highest network throughput, but did not fully deem the huge network node density; Reference [22] proposed a carrier sense multiple access (CSMA) scheme based on unidirectional handshake for terahertz networks.

(2) Fixed multiple access mainly uses frequency and time division. Reference [21] proposed the use of FTDMA technology. In different time slots, the frequency is divided into several subcarriers, and the subcarriers used by each user are orthogonal. The path loss and noise impact of the terahertz frequency band are not ignored. Reference [17] uses a multiple access mechanism based on time division multiple access (TDMA) and proposes a directional MAC protocol for terahertz communication links. It uses pulse-level beam switching and energy control, but the overhead is too large. In addition, reference [9] also discusses a channel access scheme based on TDMA in vehicle communication scenarios. This scheme uses SDN to switch between millimeter wave and terahertz bands to achieve high-bandwidth data transmission, but it also has the problem of excessive overhead.

(3) In hybrid multiple access, reference [8] uses a hybrid access scheme based on IEEE 802.15.3c protocol using carrier sense multiple access with collision avoidance (CSMA/CA) and TDMA. Information transmission is implemented by multiple superframes, each of which consists of a beacon (BP), a channel access frame (CAP), and a channel time allocation frame (CTAP). During the CAP period, channel access is performed based on CSMA/CA; during the CTAP period, time slots are allocated to network devices through TDMA.

3.3 Neighbor Node Discovery

Network nodes in terahertz networking situations must be aware of their neighbors' locations before transmitting directed data in order to fully align and beamform. Therefore, efficient network neighbor node discovery is a prerequisite for establishing terahertz communication links. So far, there are two mainstream solutions.

(1) Sub-6 GHz and terahertz fusion communications [19]

The scheme is divided into two stages: first, using sub-6 GHz omnidirectional antennas to broadcast data and obtain location information; then using terahertz directional antennas to perform beam alignment and data transmission based on location information. However, frequent switching seriously increases equipment costs and control overhead.

(2) Node scanning and discovery based on narrow beam directional antennas

Traditional exhaustive scanning schemes can lead to high latency and even communication interruption. Reference [23] uses a heuristic algorithm to propose a fast scanning algorithm suitable for millimeter wave directional beams.

4. Problem and challenges

The current research status shows that terahertz transmission characteristics are the root cause of networking problems (the relationship between terahertz transmission characteristics, networking problems, and technical challenges is shown in Figure 5). This section starts with terahertz transmission characteristics, the problems and technical challenges faced by THz wireless network, and further points out the technical development direction of breaking through the bottleneck of terahertz networking from the perspective of protocols and network strategies at all levels.

4.1 Problems

(1) High link initialization delay

In the terahertz wireless networking scenario, highly directional antennas not only bring node and channel perception problems, but also bring challenges to link initialization establishment [24]. Reference [25] used the TeraSimns-3 terahertz end-to-end network simulation tool to simulate and compare the establishment time of millimeter wave links and terahertz links in mobile network scenarios, proving the high latency of terahertz directional link establishment (the initialization latency of millimeter wave/terahertz links is shown in Figure 6). In this simulation scenario, the millimeter wave link and the terahertz link use the same 3GPP NR frame structure and time waiting model.

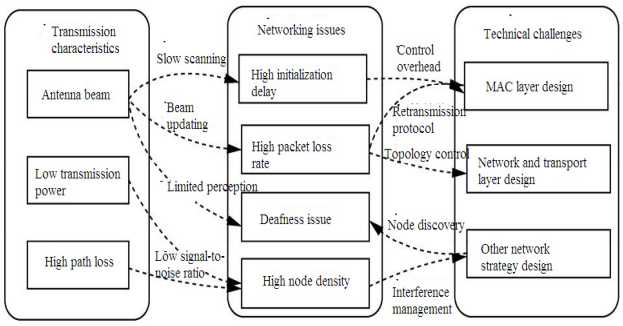

Fig. 5. Relationship between THz transmission characteristics and technical challenges

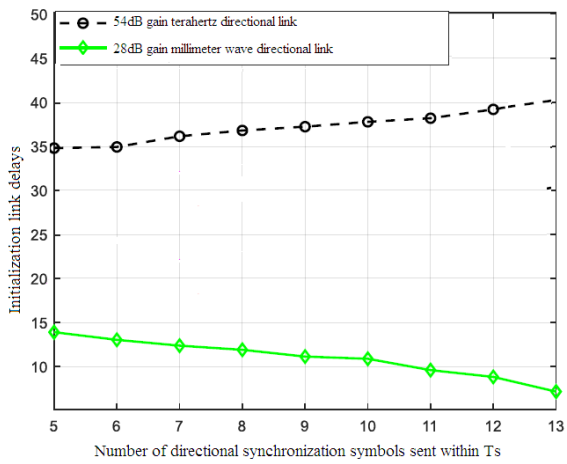

Fig. 6. Millimeter wave/THz link initialization delay[25]

(2) Link retransmission effect is poor and packet loss rate is high

The terahertz buffer queue has a limited length and will be quickly filled up when processing rates of tens or even hundreds of Gigabits per second, causing unpredictable congestion and resulting in network packet loss. In addition, the narrow beam and frequent updates of the terahertz base station will disrupt the confirmation flow between the sender and the receiver, resulting in packet transmission failure.

(3) High network deployment density, small coverage, and high energy consumption

In order to solve the network coverage problem, reference [25] constructed a Poisson point process network random point model based on Monte Carlo simulation, and compared the association among the deployment density and coverage of THz base stations and millimeter wave base stations is shown in Figure 7. If an ideal narrow beam directional antenna (1 024×256 MIMO antenna, gain 54 dB) is used to increase the link budget and improve network coverage, 0.43 THz base stations need to deploy 100 base stations/km2 to achieve 95% coverage, with an average cell radius of 56 m; while 1.5 THz base stations need to deploy 600 base stations/km2 , with an average cell radius of 23 m. In contrast, 30 GHz millimeter wave base stations only need smaller gain antennas to achieve higher coverage.

4.2 Technology Development Trends and Challenges

In view of the problems faced by terahertz networking, it is urgent to conduct research from the perspective of network protocol and strategy design to improve network throughput, coverage, latency and other performance, so as to deeply integrate terahertz communication links with existing network scenarios and achieve efficient end-to-end data transmission. This section will start from the perspective of protocol design and explain the development trends and challenges of MAC layer design, network and transport layer design, and other network strategy designs faced by terahertz wireless networking.

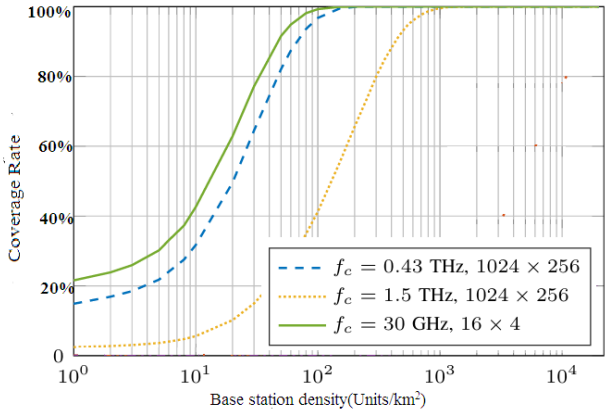

Fig. 7. Base station density vs coverage in various frequency bands[25]

4.2.1 MAC Layer Design

According to the previous discussion, the traditional MAC layer network protocol cannot solve many problems caused by narrow beam directional transmission. It is urgent to propose a scenario-oriented MAC layer protocol to achieve effective channel access, control information exchange, efficient link establishment, efficient mobility management and effective data retransmission in terahertz wireless networks. The design of MAC layer protocol for THz wireless network needs to centre on the following aspects.

(1) Design of frame structure

The duration and length of data frames are vital to network performance. For THz wireless network scenarios, the current 3GPP NR frame structure has the problem of long link establishment time, and short pilots can send more synchronization signals without affecting the control overhead, so that the tracking signals of multiple users can be multiplexed on different frequencies, thereby reducing the waiting delay for link establishment; but it will also reduce the channel estimation performance, making it more difficult to recover the frequency response of the data carrier, thereby reducing network capacity and effective coverage.

(2) Beam management protocol design

Based on the characteristics of terahertz narrow beams, in terms of beam control and beam management, in addition to breakthroughs in antenna structure, efficient beam control solutions need to be considered from the perspective of the MAC layer. In the field of millimeter waves, some beam management alternatives that are superior to exhaustive scanning have been proposed, which have certain reference significance for terahertz narrow beam management. However, the design of terahertz networking beam management solutions still faces challenges. First, the design of beam management protocols wants to judge the equilibrium among link budget and node perception. Secondly, for complex networking scenarios with multi-band fusion, it is essential to suggest multi stage beam management solutions, using beams of different widths in scenarios with different link requirements and networking requirements to achieve user tracking and channel detection. Finally, terahertz networking may need to support multi-

band MAC layer protocols to complete different network tasks, use low-frequency omnidirectional antennas for node perception, and use terahertz directional beams for data transmission. Therefore, there is an urgent need for a beam management protocol that can achieve flexible switching between omnidirectional antennas and directional antennas.

(3) Multiple access technology
For fixed multiple access schemes, its advantages are strong conflict avoidance ability, high spectrum efficiency, and adaptive tracking of links. However, the high path loss of terahertz communication links must be considered, and the control plane needs to adapt to directional antennas. Fixed multiple access faces the challenge of maintaining connectivity between network nodes. For random multiple access schemes, its advantages are high system flexibility, low complexity, and small interference conflicts between concurrent data. However, the data transmission process requires beam alignment between the receiving/transmitting ends. How to design fast node scanning and high-precision synchronization before link establishment is a challenge for random multiple access.

(4) Retransmission protocol design
According to the previous discussion, terahertz wireless networking has problems such as link retransmission difficulties and hardware breakthroughs, and it is necessary to design a more concise and effective retransmission protocol. Network coding technology does not require retransmission of specific data packets, but the receiver decodes a random combination of a batch of data packets. This technology provides a development direction for terahertz link retransmission, but the retransmission problem is still one of the challenges that need to be solved in MAC layer design.

4.2.2 Network and transport layer design
In addition to the MAC layer protocol design, the network layer and transport layer design are key technologies for terahertz wireless networking in maintaining the network structure and achieving efficient data transmission between layers. It includes the following aspects.

(1) Topology control
Section 3.1 has discussed the centralized and distributed topologies of terahertz wireless networking, both of which need to consider the issue of maintaining directional link connectivity. There are two main challenges: the terahertz network node beam updates quickly, and the antenna direction will be periodically opened and closed. Especially in mobility scenarios, it is difficult for network nodes to maintain real-time connections; terahertz wireless networking has a high deployment density, and based on energy-saving considerations, network nodes may adopt a periodic sleep mechanism, which will affect the optimal routing topology of the network.

(2) Control plane design
After comprehensively considering the terahertz path loss, base station and energy consumption costs, completely independent terahertz networking is difficult to achieve in the short term, and terahertz communication links need to be integrated with existing networks. This will use new terahertz phased arrays, nano-reflective materials and other network elements to improve link budgets and enhance the coverage of terahertz base stations, thereby reducing base station deployment density and optimizing network energy consumption. The integration of these new network elements with fixed infrastructure requires the design of dedicated control plane protocols to manage switching between users connected to different reflective devices and to enable user tracking in high-speed mobile scenarios.

(3) User plane design
In the process of realizing terahertz networking, in addition to the terahertz frequency band, it is also necessary to integrate millimeter waves and low-frequency bands below 6 GHz. Therefore, both network base stations and mobile devices need to flexibly use the frequency bands with the best performance to achieve optimal use of the spectrum. The user plane design in the terahertz wireless networking scenario needs to take into account the difficulty of multi-connectivity between user interfaces. For example, when communicating at close range, users connect to the terahertz wireless interface to achieve high-speed transmission; when the transmission distance is long, users switch to millimeter wave links, low-frequency band communication links, etc. in a timely manner.

4.2.3 Other network strategy designs

- Design of neighbor node discovery scheme
- Interference management scheme design

In the terahertz wireless networking scenario, large bandwidth and high directivity reduce interference conflicts. However, serious interference problems still occur when node beams cross and long-term data transmission occurs. In addition, ultra-dense network nodes and user mobility also aggravate the impact of interference. It is urgent to establish a new interference model that can capture the characteristics of the terahertz frequency band and track and predict the interference source.

5. Conclusion

Terahertz wireless networking is suitable for scenarios such as wireless local area networks/personal area networks, information showers, hotspot cellular coverage, Internet of Vehicles, military communications, and space communications. Research results on scenario-oriented network topologies control, Medium Access Control layer protocol design, and neighbor node detection have promoted the development of THz communication. Terahertz wireless networking still has issues with high connection latency, low retransmission efficiency, directional deafness, and excessive node density, though, because of hardware limitations. It is urgent to further study the MAC layer, network and transport layer protocol stacks for scenarios, and develop frame structure, beam organization, retransmission protocols, node discovery strategies, interference management and other technologies.

This is an Open Access article distributed under the terms of the Creative Commons Attribution License.

References

[1] P. Mugen, S. Yaohua and W. Wenbo, "Intelligent 6G wireless access network: architecture, technology and prospects," *J. Beijing Univ. Posts Telecommun.*, vol. 43, no. 3, pp. 1 – 10, Mar. 2020.

[2] X. Hongyu and Z. Xinran, "Access network architecture of 5G mobile communication," *Telecommunic. Sci*., vol 34, pp. 10-18, May 2018.

[3] X. Li, J. Yu, J. Zhang, Z. Dong, F. Li, and N. Chi, "A 400G optical wireless integration delivery system," *Opt. Express*, vol. 21, no. 16, Aug. 2013, Art. no. 18812, doi: 10.1364/OE.21.018812.

[4] Q. Wu *et al.*, "A 21 km 5 Gbps real time wireless communication system at 0.14 THz," in *2017 42nd Int. Conf. Infrar., Millimeter, Terahertz Wav. (IRMMW-THz)*, Cancun, Mexico: IEEE, Aug. 2017, pp. 1–2. doi: 10.1109/IRMMW-THz.2017.8066870.

[5] Y. He, Y. Chen, L. Zhang, S.-W. Wong, and Z. N. Chen, "An overview of terahertz antennas," *China Commun.*, vol. 17, no. 7, pp. 124–165, Jul. 2020, doi: 10.23919/J.CC.2020.07.011.

[6] M. P. Tong ZHANG Yijing REN, Shi YAN, "Artificial intelligence driven 6G networks:endogenous intelligence," *Telecommunic. Sci.*, vol. 36, no. 9, Sep. 2020, Art. no. 14, doi: 10.11959/j.issn.1000-0801.2020266.

[7] V. Petrov, D. Moltchanov, and Y. Koucheryavy, "Applicability assessment of terahertz information showers for next-generation wireless networks," in *2016 IEEE Int. Conf. Commun. (ICC)*, Kuala Lumpur, Malaysia: IEEE, May 2016, pp. 1–7. doi: 10.1109/ICC.2016.7511129.

[8] L. You, Z. Ren, C. Chen, and Y.-H. Lv, "An Improved High Throughput and Low Delay Access Protocol for Terahertz Wireless Personal Area Networks," *J. Comp.*, vol. 28, no. 3, pp. 147–158, Jun. 2017, doi: 10.3966/199115592017062803012.

[9] A. S. Cacciapuoti, K. Sankhe, M. Caleffi, and K. R. Chowdhury, "Beyond 5G: THz-Based Medium Access Protocol for Mobile Heterogeneous Networks," *IEEE Commun. Mag.*, vol. 56, no. 6, pp. 110–115, Jun. 2018, doi: 10.1109/MCOM.2018.1700924.

[10] S. Xuemin, Ch. Nan and Z. Haibo, "Air-space-ground integrated network technology: exploration and prospect," *J. Intern. Things.*, vol 4, no. 3, Mar. 2020.

[11] "China Unicom CUBE-Net 3.0 Network Innovation System." Research Institute of China United Network Communications Group Co., Ltd., Mar. 2021. [Online]. Available: https://www.bianyuanyun.com/

[12] "Intelligent vehicle networking system with integrated perception and communication for beyond-horizon perception", Key Laboratory of Pan-network Wireless Communications, Ministry of Education, Beijing University of Posts and Telecommunications, 2020.

[13] S. Thurpati and P. Muthuchidambaranathan, "Design of Hybrid Precoding for Millimeter-Wave Massive MIMO System," *Wireless Pers. Commun.*, vol. 127, no. 4, pp. 3363–3377, Dec. 2022, doi: 10.1007/s11277-022-09922-5.

[14] M. Jiao, M. Peng, and C. Liu, "Research on coverage and switching performance of UAV networks," *J. Beijing Univ. Posts Telecommun.*, vol. 43, no. 6, pp. 74–81, Jun. 2020.

[15] L. R. LU Haitao YAN Xincheng, ZHOU Qiang, DAI Jiulong, "Key Intrinsic Security Technologies in 6G Networks," *ZTE Communications*, vol. 20, no. 4, pp. 22–31, Apr. 2022, doi: https:

[16] J. Wang, C. Yang, and N. Yan, "Study on digital twin channel for the B5G and 6G communication," *Chinese J. Radio Sci.*, vol. 36, no. 3, pp. 340–348, 385, Mar. 2021, doi: 10.12265/j.cjors.2020240.

[17] J. Lin and M. A. Weitnauer, "Pulse-level beam-switching MAC with energy control in picocell Terahertz networks," in *2014 IEEE Global Communic. Conf.*, Austin, TX, USA: IEEE, Dec. 2014, pp. 4460–4465. doi: 10.1109/GLOCOM.2014.7037510.

[18] C. Han, W. Tong, and X.-W. Yao, "MA-ADM: A memory-assisted angular-division-multiplexing MAC protocol in Terahertz communication networks," *Nano Communic. Netw.*, vol. 13, pp. 51–59, Sep. 2017, doi: 10.1016/j.nancom.2017.08.001.

[19] W. Tong and C. Han, "MRA-MAC: A Multi-Radio Assisted Medium Access Control in Terahertz Communication Networks," in *GLOBECOM 2017 - 2017 IEEE Global Communic. Conf.*, Singapore: IEEE, Dec. 2017, pp. 1–6. doi: 10.1109/GLOCOM.2017.8254582.

[20] Q. Xia and J. M. Jornet, "Cross-layer analysis of optimal relaying strategies for terahertz-band communication networks," in *2017 IEEE 13th Int. Conf. Wirel. Mobile Comp., Netw. Communic. (WiMob)*, Rome: IEEE, Oct. 2017, pp. 1–8. doi: 10.1109/WiMOB.2017.8115823.

[21] Z. Li, L. Guan, C. Li, and A. Radwan, "A Secure Intelligent Spectrum Control Strategy for Future THz Mobile Heterogeneous Networks," *IEEE Commun. Mag.*, vol. 56, no. 6, pp. 116–123, Jun. 2018, doi: 10.1109/MCOM.2018.1700912.

[22] Q. Xia, Z. Hossain, M. Medley, and J. M. Jornet, "A Link-Layer Synchronization and Medium Access Control Protocol for Terahertz-Band Communication Networks," *IEEE Trans. Mobile Comput.*, vol. 20, no. 1, pp. 2–18, Jan. 2021, doi: 10.1109/TMC.2019.2940441.

[23] A. Patra, L. Simic, and M. Petrova, "Experimental evaluation of a novel fast beamsteering algorithm for link re-establishment in mm-wave indoor WLANs," in *2016 IEEE 27th Annual Int. Symp. Pers., Indoor, Mob. Radio Commun. (PIMRC)*, Valencia, Spain: IEEE, Sep. 2016, pp. 1–7. doi: 10.1109/PIMRC.2016.7794603.

[24] Q. Xia and J. M. Jornet, "Expedited Neighbor Discovery in Directional Terahertz Communication Networks Enhanced by Antenna Side-Lobe Information," *IEEE Trans. Veh. Technol.*, vol. 68, no. 8, pp. 7804–7814, Aug. 2019, doi: 10.1109/TVT.2019.2924820.

[25] M. Polese, J. M. Jornet, T. Melodia, and M. Zorzi, "Toward End-to-End, Full-Stack 6G Terahertz Networks," *IEEE Commun. Mag.*, vol. 58, no. 11, pp. 48–54, Nov. 2020, doi: 10.1109/MCOM.001.2000224.